

Question 1: Deep Learning (10 points)

- a) Suppose there is a fully connected feed forward neural network with 4 inputs, 2 outputs, and 3 hidden layers. Calculate the number of weights in this neural network if it has:

6 neurons in the first hidden layer

7 neurons in the second hidden layer

5 neurons in the third hidden layer

Note: Do not consider biases.



$$(4 \cdot 6) + (6 \cdot 7) + (7 \cdot 5) + (5 \cdot 2)$$

$$24 + 42 + 35 + 10$$

$$= \boxed{111}$$

- b) Write the formulas for the following activation functions:

Binary Step

$$f(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

Logistic (Sigmoid)

$$f(x) = \frac{1}{1 + e^{-x}}$$

ReLU (Rectified Linear Unit)

$$f(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$

Question 2: Computer Vision (10 points)

a) In a convolution layer in a CNN, if the original image size is 25×25 pixels and the kernel size is 5×5 , calculate the size of the convolved image (stride is 1 and there is no padding).

$$\left(\frac{25 - 5}{1} \right) + 1 = 21 \quad \boxed{21 \times 21}$$

b) Describe the three types of pooling operations.

- max pooling: picks the biggest val. in each region
- average pooling: takes the average of each region
- sum pooling: adds up all the vals in each region

Question 3: Natural Language Processing (10 points):

a) What are the 2 types of text summarization in NLP?

- extractive: picks out any important sentences + phrases from the original text then puts it together
- abstractive: gives new sentences that capture the same meaning of the original text

b) What is sentiment analysis in NLP?

It's how one can determine the emotional tone conveyed in some piece of text

c) In NLP, how are input texts of varying sizes converted to a fixed length for a neural network? Name and explain the two simple methods.

- Padding: a shorter text is filled extra tokens until it reaches a set max length.
- Truncating: a text that is too long has whatever tokens after a set length discarded